

ParkProof

Build Journal — the 5 days that shipped it

A chronological record of how ParkProof went from an empty repo on 16 May 2026 to a live, custom-domain, internationalised, cloud-synced PWA on 20 May 2026 — 70+ commits across 7 active build days. Each day gets a narrative, the commits that landed, and a decision-tree block at every pivot where a chosen path replaced one I'd ruled out.

Read this as a build artefact, not a sales document. It's honest about the dead ends — including the ~5 hours on 20 May I spent trying to bypass the API Gateway 30-second timeout with two architectures that failed before the third one stuck.



The arc, at a glance

Date	Theme	Headline work
16 May	Foundation	Initial commit + reminders + screenshot harness + PDF fixes
17 May	The Big Push	Live countdown · Cognito + DDB + S3 cloud sync · 5-language i18n → 9 langs · Layer 2 telemetry
18 May	Polish + Launch	Custom domain · 16-screenshot pipeline · case study · UX bug-fix sweep
19 May	Safety Gates	Pre-launch audit · paid-parking acknowledgement · no-sign-here flow
20 May	Crisis + Pivot	Accessibility gate · 30s timeout crisis → async-polling architecture · end-of-session feature
21 May	Harden + Domain	Test coverage filled (66 cases) · lambda contract tests (20) · 2 prompt fixes · DNS migration to Cloudflare · www.parkproof.com.au live

DAY 1 · Fri 16 May 2026 · Foundation

Day one is an initial-commit day. The repo was created from a working state — Sign Translator, Session Logger, basic reminders, PDF export, session history all scaffolded — and the first commit is essentially the entire app in one push: 88 files, ~17,400 lines. That's the snapshot of a previous build session getting pushed up to GitHub.

The three follow-up commits are the kind of work you do once the bones are in place: deepen the reminder picker (multi-offset with auto-disabled past chips), build a Playwright screenshot harness so the README can carry real captures, and harden the PDF export against the edge cases that broke when I tried to render sessions that had no signature yet.

Commits

98bd332	chore: initial commit — 88 files, ~17,400 lines
e6396b4	feat(reminders): multi-select offsets with auto-disabled past chips
5a236b4	feat(tooling): automated screenshot harness via Playwright
98d2f17	fix(pdf): surface export errors, guard malformed signatures, fix expired-session copy

DECISION · Where to render the parking-spot timezone — device or spot?

Q: Show '3:45 pm' in the user's device locale, or in the parking spot's local time?

```
+-- A · Device locale
|   (simpler; works without GPS)
|   ✗ Travelling Sydneysider scans in Melbourne, sees device-Sydney
|     time on a Melbourne sign. Off by an hour at DST boundaries.
|
CHOICE -+
|
+-- B · Spot timezone via tz-lookup ← CHOSEN
|   ✓ Times always match the sign's reality
|   ✓ One npm dep (tz-lookup, 6.1 MB) vs hand-rolled offset math
|   ✓ Works offline (lookup is bundled, not API)
```

DAY 2 · Sat 17 May 2026 · The Big Push

The single heaviest day of the build — 15 commits including the two largest single landings of the project: a 3,495-line cloud-sync pivot (Cognito User Pool + JWT-gated API Gateway routes + DynamoDB + S3 evidence bucket, end-to-end) and a 1,704-line internationalisation expansion (Hindi + Punjabi added on top of the five-language scaffolding from earlier the same day).

The order matters: Tier-A features (photo-quality pre-check, walk-back navigation, the home 'Currently parked' card with urgency-coloured countdown, KMS cryptographic signing) shipped *before* the auth pivot. That kept the user-facing surface complete-as-a-free-app first; cloud was layered on as opt-in durability second. Anonymous-by-default is a deliberate ParkProof design value — every feature works without a login wall.

DECISION · Cloud-sync backend — which managed identity + storage stack?

Q: How do users sign in optionally, and where does their data live?

```

      +-- Firebase Auth + Firestore
      |   X New cloud vendor, separate billing,
      |   another security surface to harden
      |
      +-- Supabase (Postgres + Auth + Storage)
      |   X Tempting (great DX), but introduces a
      |   second hyperscaler runtime in the stack
      |
Identity + storage  -----+
      |
      +-- AWS Cognito + DDB + S3 ← CHOSEN
          ✓ Same AWS account as the Lambda
          ✓ Cognito JWT authoriser bolts onto the
            existing API Gateway routes natively
          ✓ KMS signing key already lives here
          ✓ One bill, one IAM model, one CloudWatch
          ~ Cognito's Hosted UI is dated — accept it
            for federated login, custom-style email/pw
```

DECISION · Adding multi-lingual UI — locale picking strategy?

Q: Which languages to ship and how to source the list defensibly?

```

      +-- Most-spoken globally (CN, ES, AR, FR, RU...)
      |   X Generic. Doesn't match the user.
      |
      +-- Most-spoken in Australia (incl. EN regional)
      |   X Wide list, low signal per locale added.
      |
Locale set  -----+
      |
      +-- City of Melbourne LGA top non-English ← CHOSEN
          langs from 2021 ABS Census
          ✓ Direct match to actual user population
          ✓ Final 7: EN, Zhongwen (zh-CN), Tieng Viet,
            Italiano, Ellinika (el), Hindi (hi), Punjabi (pa)
          ✓ Defensible in any PM interview as a
            data-grounded scoping decision
```

Commits

a58cd4c	feat(home): live 'Currently parked' card with urgency-coloured countdown
9c0691b	feat(time): date-aware 'Until' / 'Move by' / reminder labels
402f531	feat+fix: Tier-A features + three gotcha fixes (1,053+ lines)
1d20352	docs(readme): refresh stale sections after Tier-A + gotcha shipments
c393511	feat(auth+sync): Cognito sign-in, DynamoDB + S3 cloud sync, end-to-end (3,495+ lines)
c8e7391	fix(auth): SPA boot polyfill, CORS for GET + Authorization, export as PDF
a5cd755	docs(readme): align with the now-shipped auth + PDF-export reality
3127e86	docs(readme): flag pending Apple federation
84c7d99	feat(i18n): five-language support with flag selector — EN / zh-CN / vi / it / el
d5f6f3b	docs(readme): align with multi-lingual + auth + cloud-sync reality
9542071	fix(i18n): restore inline emphasis lost when strings were flattened
d3d0419	fix: ReuseCard Italian half-translation + stale-chunk PDF export crash
6e84cff	feat(i18n): Hindi + Punjabi + PDF translations + dropdown selector (1,704+ lines)
2cdc257	docs(readme): Apple federation live
03f8006	feat(telemetry): AI feedback Layer 2 — context-aware failure analytics

DAY 3 · Sun 18 May 2026 · Polish + Launch Prep

The shape of the day changes — the deploy pipeline is humming, the feature set is broadly complete, and the brief shifts from *build features* to *make this defensible as a portfolio piece*. Twelve commits, three parallel threads: domain cutover, screenshot pipeline rebuild, and the PM case-study doc.

Bug-fix interludes throughout: the "+N more" pill on the active card was visually a tap target but technically wasn't (button-inside-button is invalid HTML and silently swallows the click); home-screen cards drifted off-centre when wrapped i18n strings pushed the layout; cross-device image sync wasn't actually working when users re-opened on a new device — root cause was S3 presign URLs minted under the wrong path prefix.

DECISION · Custom domain — registrar route?

Q: parkproof.dsouza.tech is on Network Solutions. How do we point it at CloudFront with a valid TLS cert?

```
+-- Transfer registrar to Route 53
|   X ~7-day transfer window, $ cost,
|   breaks the existing dsouza.tech setup
|
+-- Use Route 53 as DNS-only (NS records
|   delegated from Network Solutions)
|   X Adds a second DNS provider to maintain,
|   Network Solutions' NS-delegation UI is
|   known to be quirky
|
Domain → CloudFront -----+
|
+-- ACM cert in us-east-1 + CNAME on      ← CHOSEN
    Network Solutions pointing at the
    CloudFront distribution
    ✓ Single CNAME row; Network Solutions
      panel is enough
    ✓ ACM auto-renewal stays in AWS
    ✓ Cert + distro live in the same account
    X ~2 hours lost to Network Solutions
      DNS UI quirks (slow propagation,
      no DNS preview)
```

Commits

3ab2e6f	feat(domain): cut over to parkproof.dsouza.tech
79d5acc	chore(public): drop unused / off-brand orphan assets
fb1fa8b	docs: refresh README + CLAUDE for current shipped surface
7bbe469	chore: add MIT LICENSE
34cef80	docs(readme): fix stale numbers + add cloud-sync to feature list
8b6d103	fix(i18n): home-screen cards drift off-centre when translations wrap
5116c7e	feat(screenshots): expand pipeline to 16 captures + add real fixtures
558aed1	docs(screenshots): regenerate all 16 from the expanded pipeline

73df423	docs(readme): restructure Demo into 4 themed sub-grids
e8388b9	docs: add PM case study (docs/case-study.md)
b82eef6	fix(home): "+N more" pill is now a real tap target, not a UX lie
e3e1ceb	fix(sync): cross-device images now actually load

DAY 4 · Mon 19 May 2026 · Safety Gates

A quiet day by commit count (2) but big by impact. The first commit is the pre-launch audit — fixes for stale doc references, missing alt-text, OG-image metadata. The second is the day's substantive feature: **paid-parking + no-sign-here**. Both are safety mechanisms.

The paid-parking acknowledgement is the most under-appreciated decision I made on this project. Without it, ParkProof says 'you can park now' for a paid bay without the user explicitly acknowledging they need to pay — which is the exact failure mode that creates a wrongful-feeling ticket. The gate turns an implicit assumption into an explicit choice.

DECISION · Detecting 'no sign here' parking — how to capture the spot defensibly?

Q: User wants to log a park at a spot with no posted restrictions — what evidence do we capture?

```

      +-- Trust the user; save GPS + arrived_at only
      |   X Weak evidence if a council later disputes
      |       and claims a sign was always there
      |
      +-- Force the user to scan something anyway
      |   X Defeats the purpose; the spot has no sign
      |
No-sign capture  ----+
      |
      +-- Capture an *ambient surroundings* photo ← CHOSEN
          (substitutes for sign photo as visual evidence
           of 'no posted restrictions at this spot')
      + GPS + reverse-geocode + arrived_at
      ✓ Defensible photo evidence
      ✓ No AI call — no token spend, no error path
      ✓ Renders in evidence PDF with a distinct
        'No posted restrictions' banner
      X Open-ended (no expiry) — defers the
        'how do these sessions ever end?' question
        to a later day (it became Day 5's feature)
```

Commits

9fb2d45	docs: pre-launch audit fixes
11dcad4	feat: paid-parking gate + no-sign-here flow

DAY 5 · Tue 20 May 2026 · Crisis + Architecture Pivot

Today started normally and turned into a full architecture pivot. Morning: the accessibility-permit gate shipped (a parallel to yesterday's paid-parking gate — a hard RED warning when the sign requires a disability permit, blocking the Save button on an explicit acknowledgement checkbox).

Then a LinkedIn commenter publicly reported a sign that returned `{"message":"Service Unavailable"}`. The sign — a stacked Pratt Street monster (Clearway + multi-arrow + accessibility + meter zones) — was taking Claude 30-50 seconds to read carefully. API Gateway HTTP API has a hard 30-second timeout. The first commit (84ccbb1) was a band-aid: catch the 5xx, retry once, surface a friendly 'try a clearer / cropped photo' message instead of the raw AWS error.

But complex signs are exactly the kind ParkProof is supposed to help with. The band-aid was unacceptable. I spent the afternoon attempting two architectures that should have worked — both failed in ways I couldn't reasonably solve in the session window — and then pivoted to a third architecture that did work.

DECISION · Bypassing the API Gateway 30s timeout — three attempts

Q: Complex signs take 30-50s. API Gateway HTTP API caps at 30s. How?

```
+-- A · CloudFront → Lambda Function URL (OAC, sigv4)
|   Idea: CloudFront has a 60s origin-response timeout.
|       OAC signs requests to a Function URL with
|       AuthType=AWS_IAM, bypassing the gateway.
|   X Persistent sigv4 signature mismatches.
|       CloudFront's signing payload didn't match what
|       the Lambda Function URL was verifying – multiple
|       AWS forum threads describe the same dead-end.
|       Burned ~90 min. Reverted.
|
+-- B · Cognito Identity Pool + anonymous sigv4
|   Idea: Frontend gets anon AWS creds, signs Lambda
|       Function URL requests directly. No gateway.
|   X 403 from the Function URL despite IAM-simulate
|       showing the role had iam:lambda:InvokeFunctionUrl.
|       Same-account principal worked; the Cognito role's
|       signed requests didn't. Some account-level block I
|       couldn't identify. Burned ~60 min. Reverted.
|
30s ceiling --+
|
+-- C · Async-polling pipeline    ← SHIPPED
|   Idea: Stop fighting the gateway. POST returns 202 +
|       job_id immediately. Lambda self-invokes async
|       and writes the result to a DDB table with TTL.
|       Client polls GET /<route>/status/{job_id} every
|       1.5s. Each poll is a single DDB GetItem – fast,
|       fits the 30s window with room to spare.
|   ✓ Lambda's own 60s timeout is the new ceiling, not
|       API Gateway's 30s.
|   ✓ Each piece is small and well-understood (DDB TTL,
|       self-invocation, two new GET routes).
|   ✓ The Pratt Street sign now returns in ~30-45s with
|       a stepped loading state instead of a 503.
|   ~ Cost of pivoting: ~5h. Cost of NOT pivoting: a
|       permanent excuse on every complex Melbourne sign.
```

DECISION · End-of-session signal — where should the tap target live?

Q: No-sign sessions stay open-ended forever without an explicit 'I've left'.
Where do we put the button?

```

    +-- On the home card only
    |   X Hidden when the session isn't the
    |       primary active one (e.g. multi-car)
    |
    +-- Inside session detail only
    |   X Two taps from home for the common case
    |
Tap-target location  -----+
    |
    +-- Both ← CHOSEN
        ✓ Home card: pill button below walk-back
          footer, one tap from home
        ✓ Detail: full 'End session' button,
          visible for ALL active sessions
          (incl. expiry-bearing – useful when
          leaving early and you want the PDF to
          show actual duration)
```

Commits

ffef865	feat(safety): accessibility-permit gate
84ccbb1	fix(api): retry-with-friendly-error layer for 30s API Gateway timeout (band-aid)
b325a47	feat(lambda): async-job pipeline for slow Claude routes
4b0f30a	feat(api): postJsonAndPoll helper for the async-job endpoints
0255376	feat(session): driver-signalled end-of-session
3317a96	docs: document driver-signalled end-of-session + no-sign mode

DAY 6 · Wed 21 May 2026 · Hardening + Real Domain

The day after launch turns into a quiet hardening day. Two threads in parallel — test coverage from scaffolding to green, and the proper domain migration off the personal-domain subdomain (parkproof.dsouza.tech) onto a real parkproof.com.au. Live users from yesterday's LinkedIn launch keep arriving on the legacy URL the whole time; nothing breaks.

Six commits, ~1,100 lines of test code, a deferred-by-design contract-test pivot for the lambda's prompt regressions, and an ~8-step DNS / cert / CloudFront / Cognito / CORS migration to land <https://www.parkproof.com.au> as the canonical URL. Two prompt fixes (EasyPark detection, accessibility-permit gate) caught and shipped along the way.

DECISION · URL forwarding for the apex + .au variants — registrar vs Cloudflare?

Q: parkproof.com.au (apex) + parkproof.au + www.parkproof.au all need to 301-redirect to <https://www.parkproof.com.au>. Where does the redirect live?

```
+-- A · Crazy Domains URL Forwarding
|   X $26 AUD/year PER DOMAIN. ~$52/yr for the
|   two redirecting domains. Pay a registrar
|   for what's a free feature elsewhere.
|   X Reports of HTTPS forwarding flakiness
|   (cert warnings during redirect).
|
+-- B · CloudFront alt-name + edge function
|   X Requires apex-CNAME-flattening, which
|   Crazy Domains' DNS doesn't support
|   (their validator also rejects underscore
|   prefixes in CNAME values – discovered
|   mid-cert-validation).
|
Redirect host -----+
|
+-- C · Cloudflare DNS + Page Rules <-- CHOSEN
|   / Free forever (free tier supports all needed)
|   / Real 301 with valid HTTPS
|   / CNAME flattening at apex available for free
|   if we later switch to apex-canonical
|   / Free DDoS / analytics as bonus
|   / DNS imports MX/SPF/DKIM email records
|   automatically (Titan email preserved)
|   ~ ~30 min nameserver propagation (fast)
|   ~ Manual add for ACM validation CNAMEs
|   (Cloudflare's importer skips _-prefixed)
```

DECISION · Test surface — what to cover, what to defer

Q: 110 it.todo cases across 5 risk-targeted test files. Fill all of them?

```
+-- A · Fill every case end-to-end
|     X The 18 in lambda/refresh.test.js hit the live Claude
|     API. Costs ~$0.02 per scan, slow, flaky in CI, doesn't
|     test the wiring (the prompt itself is the thing under
|     test in those scenarios).
|
+-- B · Fill only the pure-function suites (66 cases)
|     / walk-back, time-format, storage all pure logic with
|     clean edge cases. ~3-4h of focused writing, all green.
|     X Leaves visible WIP in vitest output (18 todo) on the
|     lambda file. Anti-portfolio.
|
Tests -+
|
+-- C · Fill 66 pure + write contract tests for lambda <-- CHOSEN
|     / Mock @anthropic-ai/sdk, verify request SHAPE (the prompt
|     content, model config, schema enforcement, refresh-mode
|     branching).
|     / Tests catch the regressions that ACTUALLY happen:
|     someone deleting an instruction from the prompt during
|     refactor, the model name flipping, the cache_control
|     getting dropped, etc.
|     / 92 cases green + 20 contract cases = 112/112. No WIP
|     in output. Honest about what's tested vs not.
|     / Real-user behaviour on the live site IS the integration
|     test for prompt reasoning. Don't test what you can't
|     afford to fail in CI.
```

Commits

b342dac	test: fill walk-back / time-format / storage suites (66 cases)
c62adc8	test(lambda): contract tests for refresh-mode prompt + request shape (20 cases)
a2b2b4f	fix(prompt): EasyPark / payment-methods detection regression
d9437fa	fix(prompt): ☐-only bays are parkable (with permit), not a hard block
cd6130f	feat(domain): cut over canonical URL to www.parkproof.com.au

parkproof.dsouza.tech stays alive for 7 days as fallback. Old evidence PDFs continue to verify against the public key served from either domain. Total Day 6 cost: ~\$0 on AWS (Cloudflare free tier eats the DNS work), saved ~\$52 AUD/year vs. Crazy Domains' URL-forwarding upcharge.

What this build proved

1 · Sequencing > stack

Tier-A features shipped before the auth pivot. The home countdown shipped before i18n. The screenshot harness shipped before the README rewrite. Order matters more than tooling — every later piece either built on or was made cheaper by the order of what came before.

2 · Anonymous-by-default is a real design principle

Every feature works without a login. Cloud sync is opt-in, never gated. The local copy stays the source of truth even when signed in. This isn't just a UX choice — it's why the safety gates (paid-parking, accessibility-permit) and the no-sign flow could ship without an auth dependency, and why the architecture stayed simple enough that the Day 5 timeout pivot didn't have to fight Cognito's session lifecycle on top of everything else.

3 · The right pivot is the one that simplifies the system

Two architectures failed on Day 5. Both would have required deep knowledge of AWS-account-level quirks to debug. The third — async polling — required none. It decomposes into three boringly small pieces (a TTL'd DDB table, an `InvocationType: 'Event'` self-invoke, two new GET routes), all of which fit in their respective service's normal envelope. The pivot wasn't more clever; it was less clever, and that was the point.

4 · Build journals are useful in the same week

Most build retrospectives are written months later, when the dead ends have been softened into a tidy success story. The 5 hours of Day 5 dead-ends are still raw in this document because the document is being written hours after they happened. That rawness is the value — a future hiring manager (or future-me planning the next build) gets to see the trade-offs as they actually were.

Appendix · Full commit log

Every commit on main, oldest first.

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1	98bd332	16 May	chore: initial commit
2	e6396b4	16 May	feat(reminders): multi-select offsets with auto-disabled past chips
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